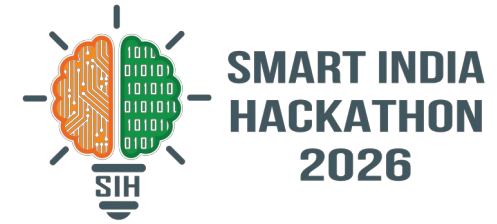


# SMART INDIA HACKATHON 2026



- **Problem Statement ID:** 26175
- **Problem Statement Title:** DepthWizard: Single-View Height Estimation & 3D Tactical Digital Twin for Rapid Reconnaissance
- **Theme:** Disaster Management
- **PS Category:** Software (AI/ML + WebGPU GIS Digital Twin)
- **Team ID:** [Insert Registered Team ID]
- **Team Name:** Control X



# DepthWizard: Single-View Height Estimation and 3D Flythrough

## OUR SOLUTION

DepthWizard automates 3D digital twins via 3 Real-Time Ingestion Streams:

- Locality Search: Direct geocoding of cities & landmarks worldwide.
- Precise Coordinates: Lat/Long decimal targeting for remote grids.
- Frontline Live GPS: 1-click automated satellite acquisition.

Monocular Depth AI yields metric DSM, flood sim & safe escape routes in < 5s.

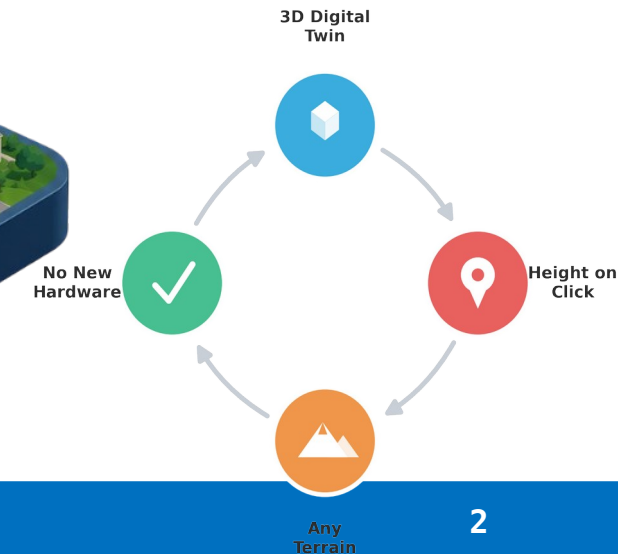
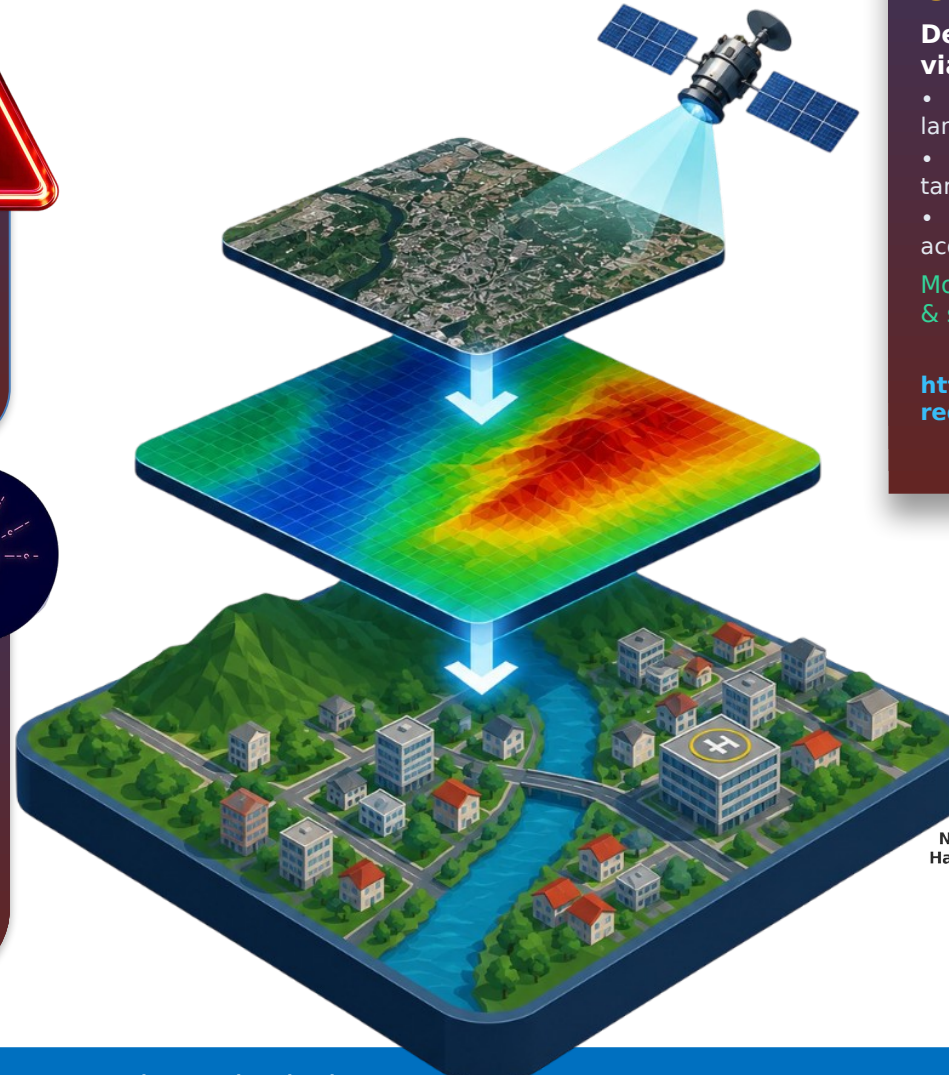
**Live Prototype Portal:**  
<https://blank-gods-prayers-reduced.trycloudflare.com>

## THE PROBLEM

Standard elevation mapping (stereo, LiDAR, InSAR) is expensive, complex, and often too slow for urgent needs like defence, urban planning, and disaster response.

## WHY IT MATTERS?

Standard monocular depth models estimate relative depth for ground-level photos, but fail on satellite imagery due to domain shift and lack of absolute scale and solving this is our core goal.



**1**  
**IMAGE /  
COORDINATE INPUT**

User uploads a PNG/JPG or GeoTIFF, or enters coordinates to auto-fetch Sentinel imagery via Copernicus.

**2**  
**MONOCULAR DEPTH ESTIMATION**

A pretrained depth model (Depth Anything V2 / MiDaS) generates a relative depth map for the image.

**3**  
**SCALE CALIBRATION**

SRTM elevation lookup (natural/georeferenced terrain) or vanishing-point + reference-object single-view metrology (man-made structures) converts relative depth into real metric height.

**4**  
**3D MESH GENERATION**

The calibrated height map and original image texture are combined into a textured .glb terrain mesh.

**5**  
**INTERACTIVE VISUALIZATION**

Three.js renders the mesh with flythrough navigation and a click-to-inspect readout of height/depth at any point.

**CORE 4-STAGE PIPELINE**

- 1. Multi-Stream Ingestion (Search/Coords/GPS)
- 2. Neural Depth-to-DSM Transformer
- 3. LOD-1 3D Building Extrusion in WebGPU
- 4. Tactical Recon: Flood Sim, Triage & Helipads

**TECHNOLOGIES USED**

**AI/DEPTH:** it leverages PyTorch, HuggingFace, Depth Anything V2, and MiDaS



**3D & Viewer:** it incorporates trimesh / Open3D, .glb export, Three.js, and a Flythrough camera path.



**Calibration:** the stack relies on OpenCV, rasterio, SRTM 30m DEM, and Single-View Metrology.



**Web & Data,** it uses FastAPI, React, Copernicus Data Space, and GeoTIFF / .npy contracts.



# FEASIBILITY AND VIABILITY



# IMPACT AND BENEFITS



## TRADITIONAL SURVEY

Field mobilization + LiDAR/stereo flight + processing — typically days to weeks per site.

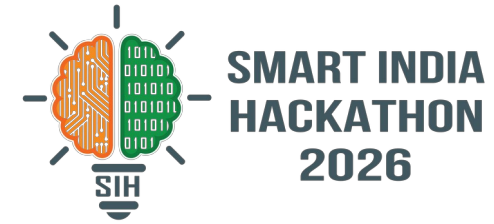
## WITH DEPTHWIZARD

One existing image, automated pipeline — elevation and a 3D model in minutes.

## SCALABILITY & QUANTIFIED ROI

- ₹0 Sensor Cost: Replaces ₹15-50L aerial LiDAR flights with existing satellite archives (98% cost reduction).
- Golden Hour Speed: Turnaround slashed from 72h to < 5s for urgent response.
- Automated Evacuation: Identifies safe havens & 3D ground escape routes.
- Aerial Heli Extraction: AI rooftop landing viability for IAF ALH Dhruv.
- National Scale: Instant deployment across NDMA, SDRF & district DEOCs.

# RESEARCH AND REFERENCES



- **GOVERNMENT & DEFENSE REGULATORY FRAMEWORKS**

1. **NDMA National Guidelines (2023):** Management of Landslides & Avalanches — High-Resolution Elevation Modeling.
2. **ISRO Bhuvan / Cartosat Framework:** Digital Elevation Models (DEM) Specifications for Disaster Monitoring.
3. **ICAO & DGCA Annex 14 Standard:** Urban Rooftop Heliport Approach Clearance Gradient ( $< 3^\circ$  roof slope).
4. **NITI Aayog Geospatial Report (2022):** Enabling Resilient Infrastructure through Open Spatial Digital Twins.
5. **IEEE Remote Sensing (2024):** Foundation Transformers for Monocular Metric Depth Estimation in Satellite Imagery.
6. **OpenStreetMap Overpass Consortium:** LOD-1 3D Vector Building Extraction & Volumetric Footprint Standards.

## LIVE EVALUATION PLATFORM & REPRODUCIBILITY LINKS

- **Live Interactive Cloud Portal:** <https://blank-gods-prayers-reduced.trycloudflare.com>
- **Official Open-Source GitHub Repository:** <https://github.com/sidharthunni/depthwizard>
- **Offline / Local Jury Evaluation Port:** <http://localhost:8080/>